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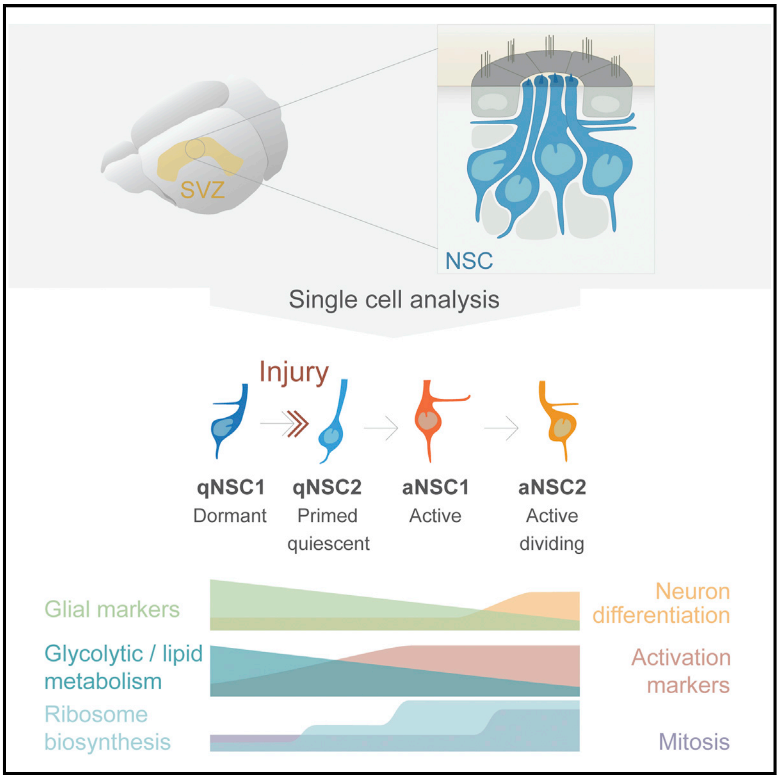
# Single-Cell Transcriptomics Reveals a Population of Dormant Neural Stem Cells that Become Activated upon Brain Injury

## Graphical Abstract

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## In Brief

Llorens-Bobadilla et al. perform single-cell RNA sequencing of acutely isolated neural stem cells (NSCs) from the adult SVZ, identifying lineage-primed NSCs residing along a continuum of co-existing states between dormancy and activation. NSCs in different states possess unique molecular features and exhibit differential responses to acute ischemic brain injury.

## Highlights

- NSCs in multiple states of activation coexist in the adult SVZ
- Dormancy is associated with high glycolytic and lipid metabolism
- Activation is associated with high protein synthesis and differentiation priming
- IFN-$\gamma$ activates a state-dependent response to acute ischemic injury in NSCs

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# Single-Cell Transcriptomics Reveals a Population of Dormant Neural Stem Cells that Become Activated upon Brain Injury

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## SUMMARY

**Heterogeneous pools of adult neural stem cells (NSCs) contribute to brain maintenance and regeneration after injury. The balance of NSC activation and quiescence, as well as the induction of lineage-specific transcription factors, may contribute to diversity of neuronal and glial fates. To identify molecular hallmarks governing these characteristics, we performed single-cell sequencing of an unbiased pool of adult subventricular zone NSCs. This analysis identified a discrete, dormant NSC subpopulation that already expresses distinct combinations of lineage-specific transcription factors during homeostasis. Dormant NSCs enter a primed-quiescent state before activation, which is accompanied by downregulation of glycolytic metabolism, Notch, and BMP signaling and a concomitant upregulation of lineage-specific transcription factors and protein synthesis. In response to brain ischemia, interferon gamma signaling induces dormant NSC subpopulations to enter the primed-quiescent state. This study unveils general principles underlying NSC activation and lineage priming and opens potential avenues for regenerative medicine in the brain.**

## INTRODUCTION

Neural stem cells (NSCs, also known as type B cells) majorly contribute to cellular plasticity in the adult brain. In the subventricular zone (SVZ) niche, these cells continuously generate neuroblasts that migrate along the rostral migratory stream to the olfactory bulb to produce new granule and periglomerular neurons (Ihrie and Alvarez-Buylla, 2011; Ming and Song, 2011).

Emerging evidence suggests that NSCs are functionally heterogeneous. For example, cells expressing different combinations of classical stem cells markers (i.e., glutamate aspartate transporter [GLAST], glial fibrillary acidic protein [GFAP], and BLBP), coexist in vivo and produce progeny with different dynamics (Calzolari et al., 2015; Codega et al., 2014; Giachino et al., 2014; Mich et al., 2014). NSCs not only differ in their state of activation, but also distinct pools of NSCs with different capacity to produce neuronal and glial progeny have been proposed to exist in the adult SVZ (Calzolari et al., 2015; Merkle et al., 2007; Ortega et al., 2013; Young et al., 2007).

However, because previous studies have been limited to the study of only a small number of factors or have been performed in potentially mixed cell populations, it is currently unknown how NSC activation and lineage determination is molecularly controlled. After brain injury, endogenous NSCs can be activated to produce multiple types of progeny to contribute to brain repair (Nakafuku and Grande, 2013; Benner et al., 2013). It is currently unclear how distinct pools of NSCs may react to brain injury and the molecular triggers of injury-induced activation of NSCs.

Here we isolated NSCs from their in vivo niche to study their molecular heterogeneity in an unbiased manner using single-cell transcriptomics. We report pervasive molecular heterogeneity in NSCs and uncover dynamic state transitions during both homeostasis and injury.

## RESULTS

### Transcriptome Analysis of Single NSCs

To resolve the molecular profiles of individual NSCs, we analyzed their transcriptome by single-cell RNA-seq. NSCs were isolated from their natural environment according to the expression of GLAST and Prominin1 (also known as CD133; Prom1). GLAST expression largely overlaps with that GFAP (Beckervordersandforth et al., 2010; Codega et al., 2014; Platel et al., 2009), and reporter expression of hGFAP-GFP together with Prom1 has been used to mark and extract SVZ NSCs (Beckervordersandforth et al., 2010; Codega et al., 2014; Mirzadeh et al., 2008). In addition, GLAST$^{+}$/Prom1$^{+}$ cells correspond to radial glia in the developing brain (Johnson et al., 2015). We first confirmed the presence of GLAST and Prominin1 in adult SVZ wholemount stainings. Consistent with their NSC identity, GLAST$^{+}$ cells coexpressed GFAP and were located at the center of the pinwheel structures (Figure 1A). In agreement with previous reports, GLAST expression colocalized with Prominin1 in some monociliated cells, but not in multiciliated ependymal cells (Figure 1A) (Lavado and Oliver, 2011). As determined by flow cytometry, CD45$^{-}$GLAST$^{+}$Prom1$^{+}$ cells constituted a distinct population that did not overlap with PSA-NCAM-expressing neuroblasts or CD24$^{hi}$Prom1$^{hi}$ ependymal cells, and only partially overlapped with O4-expressing oligodendrocytes (Figure 1C and Figure S1). To validate NSC behavior in vivo, we performed orthotopic SVZ transplants. One month after injection

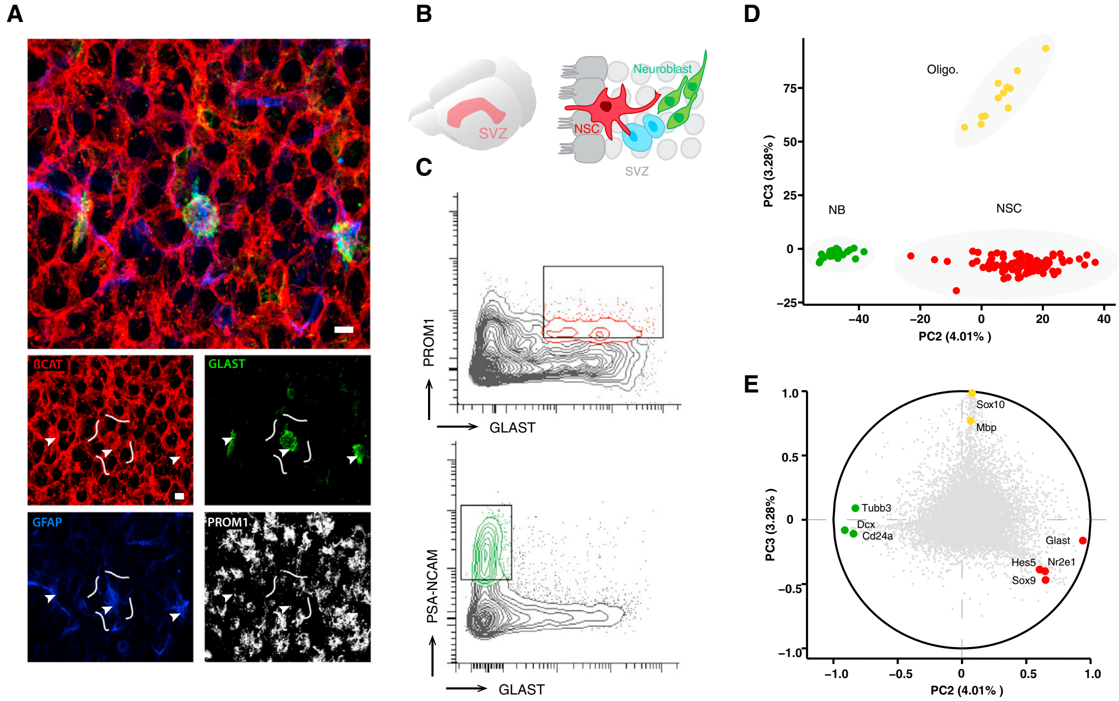


**Figure 1. Single-Cell RNA-Seq Analysis of NSCs and Neuroblasts Defines Cell-Type-Specific Transcriptomes**

(A) Confocal images of a wholemount staining of the lateral wall of the lateral ventricle. Cells were immunostained for GLAST (green), GFAP (blue), and Prominin1 (Prom-1) (white). Pinwheel structures are defined by β-catenin (red). Arrowheads mark the cilium of monociliated GLAST- and Prom1-expressing cells. Scale bar, 10 μm.

(B) Schematic representation of the SVZ.

(C) FACS strategy for the isolation of NSCs, which express both GLAST and Prom1, and neuroblasts, which express PSA-NCAM, from the SVZ niche. See also Figures S1 and S2.

(D) PCA of single-cell RNA-seq transcriptomes from 104 GLAST/Prominin1 and 26 PSA-NCAM cells. See also Table S1.

(E) Genes characterizing each population based on the correlations with principal components 2 and 3 from the PCA (each point corresponds to a gene and previously known marker genes are highlighted in color).

into the SVZ, GLAST+/Prom1+-cells expressed GFAP$^+$ and were located in the host niche from where they continuously produced olfactory bulb granule and periglomerular neurons for a period of at least 1 month (Figure S1). Thus, our combination of surface markers allows the isolation of NSCs by flow cytometry, circumventing the need for transgenic animals.

We next analyzed the transcriptome of 104 GLAST$^+$/Prom1$^+$ and 26 PSA-NCAM$^+$ cells by single-cell RNA-seq using Smart-seq2 technology (Picelli et al., 2013). Two replicates from 1,000 cells were additionally prepared as population controls. Single-cell libraries were sequenced at an average depth of 8.3 million read pairs, with an average transcriptome alignment rate of 59% (Table S1). While gene expression levels were highly correlated between population replicates (Pearson r = 0.96), we observed higher variability between single cells (Pearson 0.11 < r < 0.64; mean 0.38). Yet, the gene expression levels of the single cell ensemble highly correlate to the population samples (Pearson r = 0.85) (Figure S2). Consistent with previous studies (Treutlein et al., 2014; Wu et al., 2014), RNA spike-in standards that were included to control for technical noise showed high correlation across transcript abundances that spanned approximately five orders of magnitude (Figure S2).

We next performed principal component analysis (PCA) on all cells using the gene expression values of genes that were detected in at least two cells (TMM-FPKM > 1, 15,846 genes). This unbiased analysis clearly discriminated three main clusters of cells (Figure 1D). These clusters were assigned to different cell types after we scrutinized the genes exhibiting the highest coordinates in the respective principal component for known marker genes (Figure 1E). One cell cluster encompassed PSA-NCAM$^+$ cells and associated with expression of neuroblast markers such as doublecortin (*Dcx*), *CD24a*, and *Tubb3*. The second cluster, containing the majority of the GLAST$^+$/Prom1$^+$ cells (92 out of 104), associated with expression of NSC markers like Glast (*Slc1a3*), Tlx (*Nr2e1*), *Sox9*, *Vcam1*, and *Hes5*. The smallest cluster associated with the oligodendroglial genes *Sox10*, *Plp1*, and *Mbp* (12 out of 104) (see Figure S1). Oligodendrocytes express very low levels of GLAST and Prom1, and for further experiments we excluded them by co-staining cells with O4$^+$ in flow cytometry (see Figure S1). Thus, PSA-NCAM+ and GLAST$^+$/Prom1$^+$ cells have distinct transcriptomes that can be classified by single-cell RNA-seq and correspond to neuroblasts and NSCs, respectively.

## NSCs Are Molecularly Heterogeneous

To gain further insights into the molecular heterogeneity of these cell types, genes with the highest coordinates in the first four principal components (1,844 genes with correlation coefficient >0.5

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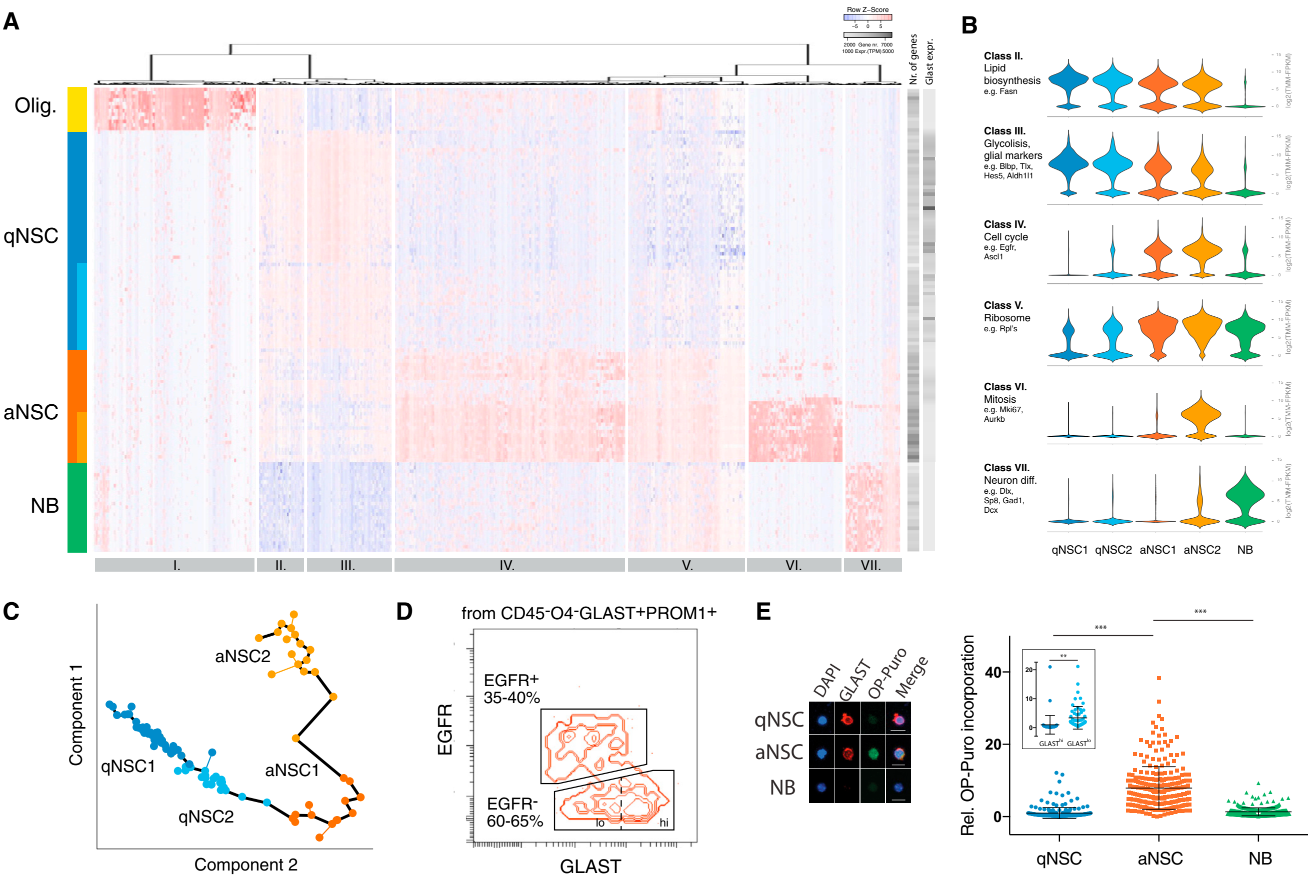


**Figure 2. Single-Cell RNA-Seq Uncovers NSC Heterogeneity**

(A) Hierarchical clustering of 1,844 genes with high correlation ($p < 1e-9$) with the first four principal components of PCA identifies four main, molecularly distinct populations. Quiescent and active NSC groups are further subdivided in two subgroups. See also Figure S3. Cells are shown in rows and genes in columns.

(B) Violin plots showing the course of expression for genes in clusters II–VII in cells from the neurogenic lineage (NSCs and neuroblasts). Main Gene Ontology terms for each class are highlighted (see also Table S2).

(C) Minimum spanning tree (MST) constructed by Monocle based on the expression pattern of 2,688 genes with the highest coordinates ($p < 10e-5$) with first two dimensions of PCA on qNSCs and aNSCs. Each point represents a cell and the solid black line indicates the pseudotime ordering. Different groups are color-coded based on PCA clustering.

(D) FACS strategy to enrich for qNSCs and aNSCs by flow cytometry based on EGFR expression. For the inset in (E), EGFR$^-$ cells were split according to GLAST expression as depicted with the dashed line.

(E) Representative image of a qNSC, aNSC, and neuroblasts (NBs) incubated with OP-Puro for 1 hr and immunostained for GLAST. Quantification of the relative OP-Puro incorporation in the different cell groups is shown at right (qNSCs = 1 ± 0.09; aNSCs = 7.91 ± 0.35; NBs = 1.35 ± 0.05; mean and SD are shown; n = 1,029 cells from four replicates; ***$p < 0.001$ by two-way ANOVA, Holm-Sidak-corrected). The inset shows quantification of OP-Puro incorporation in GLAST$^{hi}$ and GLAST$^{lo}$ qNSCs as measured 1 hr after OP-Puro and 24 hr after sorting (hi = 1 ± 0.45, lo = 3.4 ± 0.50; n = 108 cells from two replicates; $p < 0.01$; Student's t test). See also Figure S4. Scale bar, 10 μm.

and $p < 10e-9$, Table S2) were used for unsupervised hierarchical clustering. This approach further partitioned NSCs, but not neuroblasts or oligodendrocytes, into two additional subpopulations (Figure 2A). One NSC cluster associated with expression of *Egfr*, a known marker of active NSCs (Codega et al., 2014), and with cell-cycle-related genes (e.g., *Mki67* and *Mcm2*), on the basis of which we named them active NSCs (aNSCs). Conversely, the cluster that lacked activation markers was named quiescent NSCs (qNSCs). Cell-to-cell Pearson correlation coefficients, which measure the similarity between gene expression across individual cells, also validated the presence of these clusters (Figure S3). In addition, we confirmed the identity of these clusters by comparing our data to that of previous datasets (Figure S3).

To obtain functional insights into the gene classes that define each cell subtype, we performed gene ontology and pathway analyses (Huang et al., 2009) on the seven modules of co-regulated genes obtained by hierarchical clustering (Figure 2A). Of note, each cluster contains not only coding genes that may include novel marker genes, but also lncRNAs expressed in a cell-type- and state-specific manner (Table S2). As expected, genes that were uniquely expressed in oligodendrocytes, were enriched in functional categories such as myelination, synaptic transmission, and oligodendrocyte differentiation (class I). The other gene modules (class II–VII) corresponded to dynamically regulated genes between qNSCs, aNSCs, and neuroblasts (Figure 2B), likely reflecting state transitions along the neurogenic lineage. Expression of these genes further divided quiescent and active NSCs into two subpopulations each (qNSC1/2 and aNSC1/2) that reflect four different states within NSCs that have not been described previously. To gain further resolution of this partition, we performed pseudotemporal ordering of the transcriptional dynamics in these cells (Trapnell et al., 2014). Applying this algorithm, cells are ordered in "pseudotime," a quantitative measure of the progress along a biological process (NSC exit from dormancy in this case). This analysis placed qNSC1 at one end of the continuum, followed by qNSC2, aNSC1, and aNSC2 (Figure 2C). Lineage tracing of Glast-expressing cells has shown that after temozolomide-induced ablation of Glast$^{low}$ active cells, the Glast$^{high}$-subpopulation, which contains our qNSC1, generate active progeny as well as neuroblasts (Mich et al., 2014). Thus, the pseudotime analysis together with previous lineage tracing of Glast-cells suggests that a dormant state (qNSC1) precedes a primed-quiescent state (qNSC2) before an NSC becomes active.

Genes expressed in all NSC subpopulations (class II and III) have functions in metabolic pathways. Class II was associated with fatty acid metabolism and lipid biosynthesis (e.g., *Fasn* and *Ldhb*). Lipogenesis has recently emerged as a key metabolic pathway in hippocampal NSC maintenance (Knobloch et al., 2013). Our data show that this process operates in the SVZ in both quiescent and active NSCs and not in their differentiated progeny. Class III genes, which include many NSC/astrocyte markers (e.g., *Nr2e1*, *Hes5*, *Fabp7* [Blbp], *Aldh1l1*, *Slc1a3*, and *Slc1a2*), were enriched for genes involved in glycolysis. We find that expression of these genes is highest in qNSCs and already lower in aNSCs, suggesting that glycolytic metabolism is a defining feature of qNSCs, similar to quiescent stem cells in the hematopoietic system (Simsek et al., 2010). The decrease of glial-associated genes in aNSCs compared to qNSCs suggests that the glial gene-expression program starts to be shut down at the aNSC level, presumably to initiate the acquisition of neuronal fate.

The gene signature most closely associated with aNSCs comprised three gene modules functionally related to cell cycle (class IV; including *Egfr* and *Cdk2*), protein synthesis (class V; e.g., *Rpl* family), and mitosis (class VI; e.g., *Aurkb*), with this latter module separating non-mitotic-aNSC1 and mitotic-aNSC2. Genes involved in protein synthesis also explained the separation between dormant (qNSC1) and primed-quiescent (qNSC2) cells, suggesting that translational activation is one of the earlier events in the exit from dormancy. In support of this finding, PCA performed exclusively on qNSCs as well as other independent clustering methods again distinguished the same partition (Figure S3). The requirement of a tight regulation of global protein translation in stem cell function has been previously reported for hematopoietic stem cells (Signer et al., 2014). For functional validation of the RNA-seq data, we measured the rate of protein synthesis in aNSCs and qNSCs by O-propargyl-puromycin (OP-Puro) incorporation (Liu et al., 2012) directly after flow cytometry. aNSCs exhibited a 7.9-fold increase in the rate of protein synthesis as compared to qNSCs (Figures 2D and 2E and Figure S4). Consistently, after we injected OP-Puro intraventricularly, we found that Ki67$^{+}$ aNSCs are the cells with the highest translational activity in the intact SVZ in vivo (Figure S4). Notably, ependymal cells appeared almost unstained, indicating a comparatively very low level of translation (Figure S4). Because *Glast* mRNA expression is higher in dormant NSCs, we hypothesized that surface-protein expression of GLAST in combination with Prom1 and the absence of EGFR could be used to enrich for dormant and primed-quiescent NSCs (Figure 2D). Analysis of OP-Puro incorporation in GLAST$^{hi}$ and GLAST$^{lo}$ qNSCs showed that cells in the primed-quiescent state, GLAST$^{lo}$, have a higher rate of protein synthesis than their dormant counterparts, GLAST$^{hi}$ cells (Figure 2E). Together, these data validate our strategy to enrich for dormant and primed-quiescent NSCs and show that the global rate of protein synthesis defines these two states.

The last gene module comprised genes expressed in neuroblasts (class VII; e.g., *Dlx* family, *Sp8*, *Dcx*, and *Gad1*), which were functionally involved in neuron differentiation. Of note, many neuroblast-associated genes started to be upregulated at the aNSC level, indicating that these cells had initiated the differentiation program. Altogether, the transcriptome of NSCs is spread across a continuum of dormancy to activation that is associated with dynamic and stage-specific molecular signatures.

## NSCs Have a Distinct Molecular Profile

NSCs, and specially qNSCs, share many markers (i.e., *Aldh1l1* and *Gjb6*) and molecular features (i.e., high glycolytic activity) with more mature parenchymal astrocytes. This prompted us to validate the in vivo neurogenic activity of qNSCs by orthotopic transplantation. Similar to the total NSC population and as previously described (Codega et al., 2014), qNSCs entered activation in vivo and produced neuroblasts and olfactory neurons for a period of at least 1 month after being injected into the SVZ (Figure S1). We next performed single-cell RNA-seq analyses of 21 GLAST$^{hi}$ (CD45$^{-}$,O4$^{-}$) astrocytes from the striatum and the somatosensory cortex to compare their transcriptome to that of SVZ NSCs. PCA placed parenchymal astrocytes at one end of the NSC-"activation-state" continuum, preceding, and with only a minor overlap with, dormant qNSC1 cells (Figure 3A). As recently described (Zeisel et al., 2015), even at these low cell numbers, we were able to observe two main clusters of cortical astrocytes (not shown). We next set out to identify genes that would distinguish qNSC1 from parenchymal astrocytes by using a likelihood-ratio test (McDavid et al., 2013), which has been recently applied to single-cell transcriptomes of closely related cell populations (Macosko et al., 2015). This analysis revealed that qNSC1 cells express higher levels of transcripts known to be enriched in SVZ NSCs like *Thbs4* or *Rarres2* (Beckervordersandforth et al., 2010; Benner et al., 2013). We also describe the expression of additional factors that exhibit preferential expression in the SVZ in the adult brain (Figure 3B and Figure S3H). Among NSC-enriched transcripts (cluster III in Figure 2A), we found that the tetraspanins *Cd9* and *Cd81* were more highly expressed in qNSC1 than in parenchymal astrocytes (Figure 3B and Figure S3H). To validate whether CD9 expression could indeed distinguish NSCs from parenchymal astrocytes, we tested CD9 expression by flow cytometry. Consistent with the RNA-seq data, NSCs expressed substantially higher levels of CD9 in their surfaces as compared to cortical and striatal astrocytes (Figure 3C and Figure S3). To examine the spatial location of CD9 within the SVZ, we performed immunohistochemical

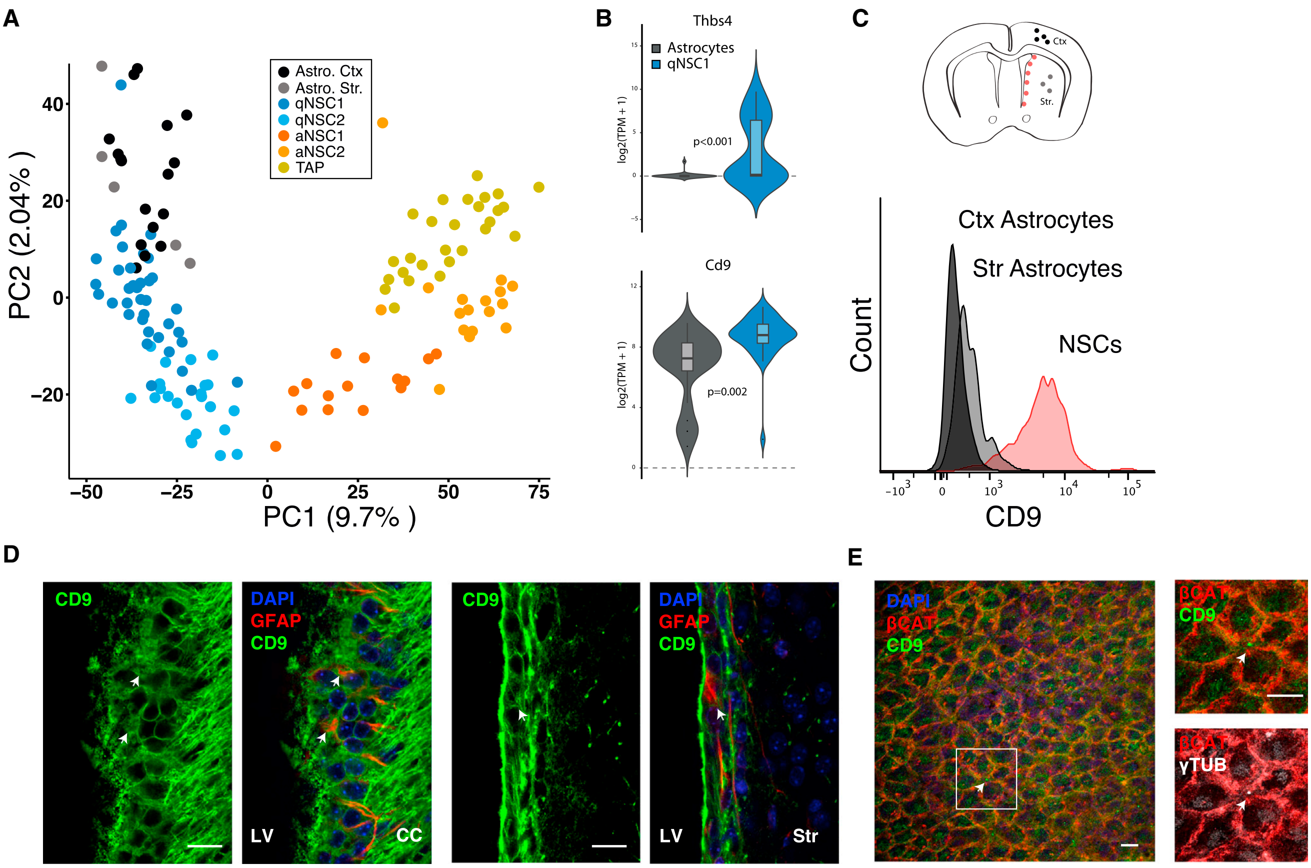


**Figure 3. Unique Molecular Properties of NSCs**

(A) PCA of the transcriptome of 92 NSCs, 21 astrocytes, and 27 TAPs. Note that parenchymal astrocytes (Astro.) and transient amplifying progenitors (TAPs) are each at one end of the NSC activation continuum.

(B) Violin plots with superimposed boxplots for the expression of *Thbs4* and *Cd9* in astrocytes and dormant NSCs (qNSC1). Benjamini-Hochberg corrected p values are by likelihood-ratio test.

(C) Scheme of the location of the cells analyzed by FACS. In the bottom, a histogram depicting both CD9 expression in GLAST$^+$ (CD45$^-$,O4$^-$) cells from the cortex (Ctx) and striatum (Str.) and GLAST$^+$PROM1$^+$ (CD45$^-$,PSA-NCAM$^-$,O4$^-$) cells from the SVZ (NSCs) is shown.

(D) Confocal image of the anterior septal (left) and lateral (right) ventricle walls immunostained for CD9. See also Figure S3. Arrows indicate the location of ventricle-contacting GFAP-expressing cells.

(E) Confocal stack of an SVZ wholemount immunostained for CD9. The first 1 μm from the ventricle is shown in the confocal stack. The arrow indicates CD9 signal at the center of a representative pinwheel. Scale bar, 10 μm. LV, lateral ventricle; CC, corpus callosum.

staining in SVZ wholemounts and coronal sections. Coronal sections showed CD9 to be highly expressed across the entire ventricular wall, which includes expression in ependymal cells and GFAP-NSCs. In the underlying parenchyma, CD9 is expressed in white matter, labeling some axons (Figure 3D and Figure S3). Wholemount staining confirmed CD9 localization across the ventricular surface, including in the membrane of some monociliated NSCs at the center of pinwheels (Figure 3E). Since aNSCs also share many molecular features (i.e., *Ascl1* expression) with transit amplifying progenitors (TAPs, also known as type C), we also analyzed the transcriptome of 27 EGFR$^+$ (CD45$^-$,O4$^-$,PSA-NCAM$^-$,GLAST$^{-/}$Prom1$^-$) TAPs. As shown in the PCA in Figure 3A, TAPs constitute a cluster that immediately follows aNSC2 along the lineage progression continuum. Altogether, NSCs in their multiple states have distinct molecular hallmarks that allow them to be distinguished from closely related cell types.

## Transcriptional Control of NSC Activation

To uncover putative transcriptional regulators of the quiescent and active states, we subjected the 92 NSC transcriptomes to PCA using only the expression values of transcription factors (TFs) (Zhang et al., 2012). Indeed, TF expression alone identified qNSC and aNSC clusters, which completely overlapped with the whole-transcriptome PCA-based separation (Figure 4A). The TFs *Sox9*, *Id2*, and *Id3*, among others, were associated with quiescence. The active state was associated with the expression of immediate early genes *Egr1*, *Fos*, SoxC factors (*Sox4* and *Sox11*), and *Ascl1*. Further, comparison of the aNSC signature (genes in classes IV–VI in Figure 2A) to recent Ascl1 ChIP-seq data (Andersen et al., 2014) revealed that a significant proportion of the aNSC transcriptome is under the direct control of Ascl1 (17.4%, $p < 1.65e{-}13$; Figure S3). These data implicate Ascl1 as a key factor in the activation of NSCs, consistent with its function in the dentate gyrus (Andersen et al., 2014). Notably, we also

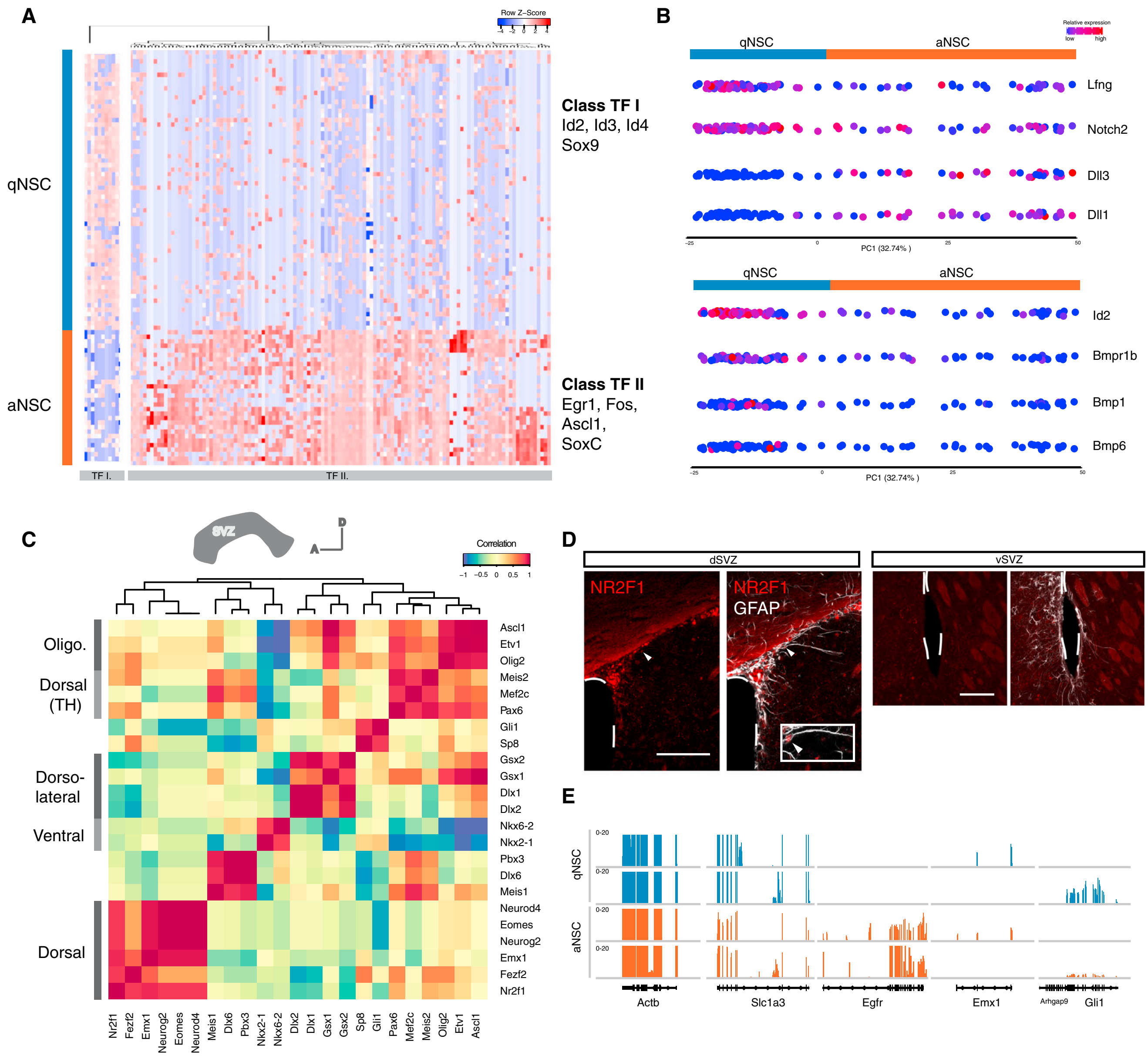


**Figure 4. Transcriptional Control of NSC Activation and Lineage Priming**

(A) Hierarchical clustering of the expression levels of 135 transcription factors with strong correlation ($p < 1e-5$) in the first two principal components in the PCA. PCA was performed on 92 NSC transcriptomes. Selected TFs in each cluster are highlighted. Cells are shown in rows and genes in columns.

(B) One-dimensional PCA of 92 NSCs. Each point corresponds to one cell and the color on each cell corresponds the scaled expression value of the respective gene in each single cell.

(C) Correlation heatmap for 23 selected transcription factors in 92 NSCs. See also Table S3.

(D) Confocal image of the SVZ immunostained for NR2F1 and GFAP. Scale bar, 100 μm.

(E) Representative genome tracks showing the expression of dorsal marker *Emx1* and ventral marker *Gli1* in two qNSCs and two aNSCs. The aNSC marker *Egfr* and two widely expressed genes (*Actb* and *Slc1a3*) are shown.

observed that neuroblast-associated TFs such as *Dlx1*, *Dlx2*, and *Bcl11a* were detected in aNSCs. Since previous cell-population-based studies could not exclude minor neuroblast contaminations (Beckervordersandforth et al., 2010; Codega et al., 2014), our single-cell approach provides now definitive evidence for the initiation of a lineage differentiation program in aNSCs based on the co-expression of stem cell- and differentiation-associated factors.

Notch signaling has been proposed as an essential regulator of quiescence and as a generator of heterogeneity in NSCs (Basak et al., 2012; Giachino and Taylor, 2014; Imayoshi et al., 2010; Kawaguchi et al., 2013). Consistently, using one-dimensional PC projections (Durruthy-Durruthy et al., 2014), we found that *Notch2* receptor and the Notch target gene *Lfng* were enriched in qNSCs (Figure 4B). Notch ligands *Dll1* and *Dll3* showed the opposite pattern of expression, suggesting that they are already segregated at the RNA level. Since quiescent and active NSCs are in direct contact in the niche (Codega et al., 2014; Kawaguchi et al., 2013; Mich et al., 2014), our data are in agreement with a model in which Dll-expressing aNSCs directly control dormancy of qNSCs. To test whether Notch inhibition would be sufficient to drive qNSCs into activation, we treated qNSCs purified by flow cytometry (GLAST$^+$/Prom1$^+$/EGFR$^-$) with the $\gamma$-secretase inhibitor DAPT. Indeed, a short DAPT treatment increased OP-Puro incorporation in qNSCs and increased the percentage of activated cells (Figure S4). We also examined the distribution of components of the BMP pathway. This pathway is known to promote quiescence in hair follicle stem cells (Genander et al., 2014). Here we found that expression of the BMP-target Id2 correlated with the expression of the receptor *Bmpr1b* in qNSCs. Interestingly, the ligands *Bmp1* and *Bmp6* followed the same pattern (Figure 4B). These results suggest that BMP, unlike Notch signaling, may regulate quiescence in an autocrine manner.

## NSCs Are Primed for Lineage Differentiation

NSCs in the adult SVZ generate several subtypes of olfactory bulb interneurons. These subtypes are believed to be derived from distinctly located sets of committed NSCs, and not from a pan-subtype multipotent NSC (Brill et al., 2009; Kelsch et al., 2007; Merkle et al., 2007; Young et al., 2007). Different combinations of TFs have been identified to be necessary or correlate with the generation of specific neuronal subtypes (reviewed in Alvarez-Buylla et al., 2008; Sequerra, 2014). The analyses of TF combinations have been limited to a handful of factors assayed at a time, precluding in-depth knowledge of the establishment and extent of this patterning. To identify the different TF combinations across the 92 NSC transcriptomes, we manually curated a list of subtype-associated TFs (Table S3) and measured their expression correlation coefficients (Shalek et al., 2013). This analysis uncovered several clusters of factors whose expression varied in a correlated manner in the first five principal components (50% of variation; Figure 4C). The presence of TFs with known spatial distribution within the clusters facilitated the reconstruction of the original location of the different NSCs within the SVZ. Dorsal markers like *Emx1* correlated with *Eomes* (also known as Tbr2), *Neurod4*, and *Neurog2*, known to specify glutamatergic interneurons in the dorsal aspect of the SVZ (Brill et al., 2009). These genes also clustered with *Fezf2* and *Nr2f1* (also known as COUP-TF1), suggesting that these genes might be involved in the specification of dorsally generated interneurons. Indeed, as predicted in silico, we found Nr2f1 expression to be dorsally restricted (Figure 4D). Ventral markers *Nkx2-1* and *Nkx6-2* were highly correlated to one another and inversely correlated with dorsal markers. Unexpectedly, they did not cluster with *Gli1*, another ventral marker (Ihrie et al., 2011), suggesting that they constitute non-overlapping domains within the ventral SVZ. Laterally expressed genes, i.e., *Gsx2* and *Dlx2*, clustered in a distinct domain, consistent with a recent report (López-Juárez et al., 2013). *Pax6* expression defined another cluster, which included its cofactor Meis2 (Agoston et al., 2014), and presumably defined the dopaminergic periglomerular neuron-generating domain. Last, we observed a cluster that included the oligodendroglial TF *Olig2*. This cluster contained *Ascl1*, which due to its broader expression was also associated with other domains (e.g., *Pax6*). Since *Ascl1*-lineage cells within the SVZ generate oligodendrocytes (Parras et al., 2004), we hypothesized that this domain corresponds to a distinct non-neurogenic NSC pool fated to become oligodendroglia (Ortega et al., 2013). Taken together, unbiased clustering of lineage-related TFs defines another layer of heterogeneity within NSCs that correlates with specific neuronal and glial subtypes in their progeny, thus highlighting the fate-restricted nature of adult NSCs. Importantly, this fate restriction is observed in both active and quiescent cells (Figure 4E).

## Ischemic Brain Injury Activates Dormant NSCs via Interferon Gamma

The analysis of NSCs during homeostasis revealed pervasive heterogeneity within NSCs. To study the transcriptional response of NSCs to a physiological perturbation in vivo, we subjected animals to a transient bilateral common carotid artery occlusion that results in ischemic striatal damage (Yoshioka et al., 2011). Single-cell RNA-seq libraries were generated from 45 NSCs isolated from the SVZ 2 days after ischemia (Figure 5A). In addition, two replicates from 1,000 NSCs likewise isolated from SVZ 2 days after ischemia were used as population controls. Combined PCA of injured and non-injured NSCs revealed that ischemic injury activated the transition of NSCs into activation (Figure 5B). Notably, we observed that only a few injured NSCs were in a dormant state while the proportion of primed-quiescent and active NSCs strikingly increased upon injury (qNSC1: 6.67% injured versus 40.2% naive; qNSC2: 37.8% injured versus 26.1% naive; aNSC: 55.5% injured versus 33.7% naive; $p = 8e{-}5$). We also noted that several injured NSCs with a primed-quiescent signature retained expression of genes otherwise only expressed by dormant cells under homeostatic conditions (Figure 5C). We hypothesize that these genes are a remnant of their previous dormant state and call them ‘‘dormancy trace genes.’’ The transition from dormant to a primed-quiescent state (qNSC1 to qNSC2) involves the activation of protein synthesis and cell-cycle genes, which is reminiscent of the alert state of quiescent muscle stem cells when exposed to a remote injury (Rodgers et al., 2014).

To identify the underlying regulatory networks, we next performed PCA on the injured NSC transcriptomes using injury-induced genes extracted from the population RNA-seq data (‘‘Injury signature;’’ 399 differentially expressed genes with 2-fold change and $p_{adj} < 0.1$, Table S4). This analysis showed that the response was not homogeneous across all NSCs, but instead revealed the presence of subsets of cells with low and high induction of the injury signature (Figure S5). We then performed unsupervised hierarchical clustering on 61 genes with the highest PC1 coordinates (correlation coefficient > 0.45 and $p < 1e{-}2$) (Figure S5). This analysis further confirmed the heterogeneous injury response in NSCs. Interestingly, most of the cells with a higher induction of the injury signature belonged to the quiescent group, while most of the aNSCs showed only a mild induction. These results also highlight that differentially expressed genes identified in population analyses may reflect high levels of induction in only certain subsets of cells within that population. Finally, we performed upstream regulator analysis using the software Ingenuity to identify putative molecular triggers of the injury response. This analysis showed an

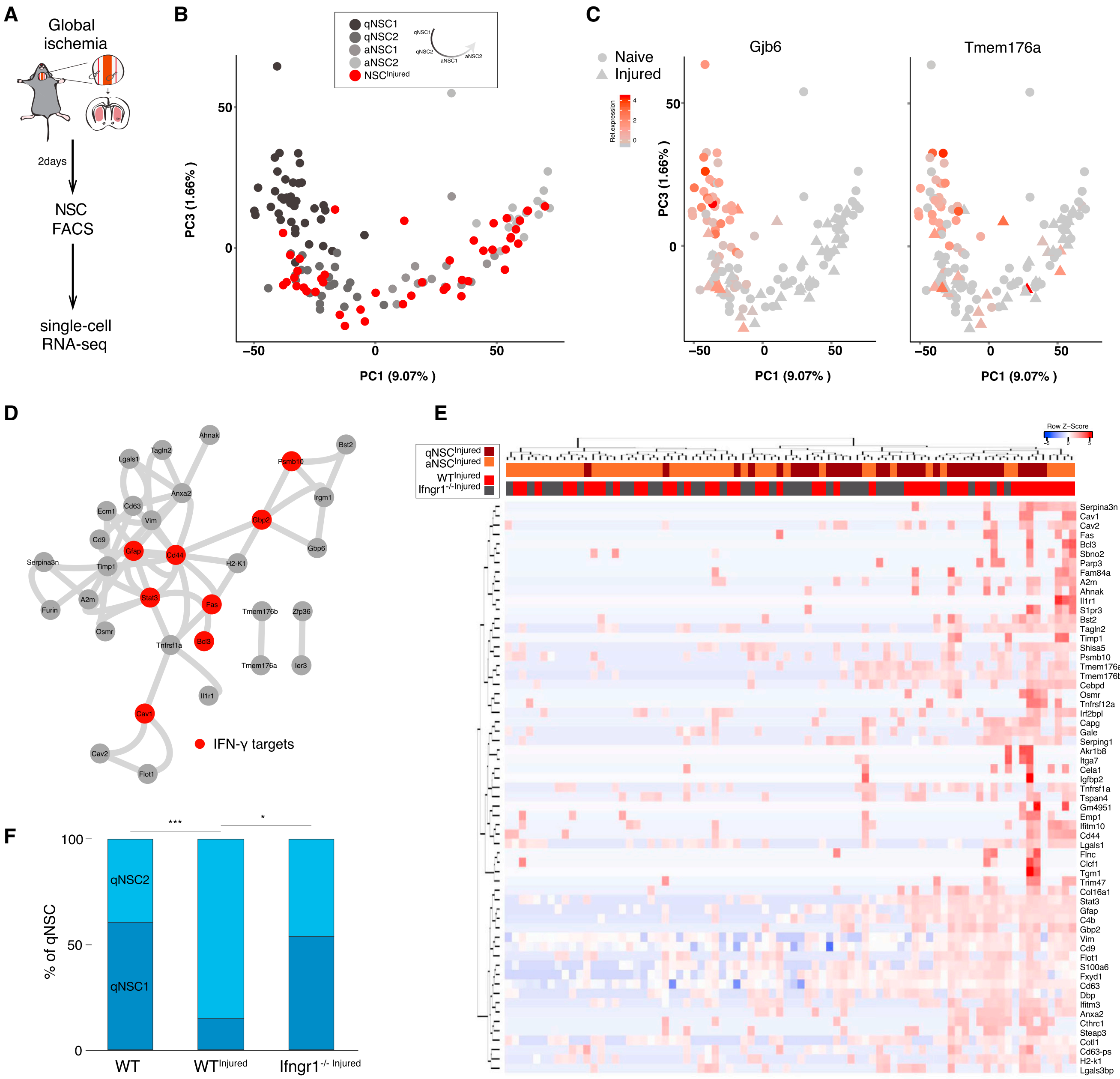


**Figure 5. IFN-γ-Driven Activation of Dormant NSCs after Ischemic Injury**

(A) Scheme of the experimental design.

(B) PCA of 124 NSC transcriptomes (92 non-injured, in grayscale; 45 ischemic injured, in red). The proportion of NSCs within the different transition states is significantly changed by the injury ($p < 0.0006$, Fisher test).

(C) Relative gene expression values of dormant-NSC (qNSC1)-associated genes projected into the PCA shown in (B). Note that some injured cells (triangles) within the qNSC2 cluster retain higher expression than their non-injured counterparts.

(D) STRING-based interaction plot for genes with the highest PC1 correlation in Figure S5A. See also Table S4. Only genes with at least one interaction are shown. In red, predicted IFN-γ target genes based on upstream regulator analysis (IPA) ($p < 4.74e{-}10$) are shown.

(E) Heatmap showing relative gene expression values across individual cells for 61 genes with the highest PC1 correlation ($p < 1e{-}2$). Genes are shown in rows and cells from WT-injured and *Ifngr1*$^{-/-}$ Injured mice are shown in columns.

(F) Relative proportions of cells (from WT, WT-Injured, and *Ifngr1*$^{-/-}$ Injured mice) in each of the qNSC subclasses as defined in Figure 2. p values (Fisher's exact test) are ***p = 0.0006 and *p = 0.02.

overrepresentation of interferon gamma (IFN-γ) target genes within our signature (Figure 5D). We next examined the transcriptome of 34 NSCs isolated from the SVZ of IFN-γ-Receptor1-deficient (*Ifngr1*$^{-/-}$) mice 2 days after ischemic injury. Indeed, the injury-induced upregulation of INF-γ-response genes was abrogated in ∼90% of the *Ifngr1*$^{-/-}$ cells (Figure S5 and Figure 5E).

Importantly, the exit of NSCs from the dormant into the primed-quiescent state drastically decreased in the absence of IFN$\gamma$R1 (60.7% qNSC1 in naive WT mice; 15.0% in injured WT mice; and 53.9% in *Ignr1*$^{-/-}$ injured mice; $p < 0.006$ and $p < 0.02$, Figure 5F). This effect was likely not due to developmental deficits in these mice since WT and *Ifnr*$^{-/-}$ mice showed similar numbers of dormant and quiescent NSCs (Figure S5).

Taken together, our data indicate that injury activates the transition of dormant NSCs into a primed quiescent state that subsequently proceeds to activation via triggering of interferon signaling.

## DISCUSSION

Despite intensive investigation on NSC identity, it has remained unclear to what degree cells in different states of activation or lineage priming exist in vivo. By applying single-cell transcriptomics to an unbiased pool of NSCs, we have reconstructed at high resolution the molecular events orchestrating NSC activation, from dormancy to the generation of diverse progeny. In agreement with previous studies (Codega et al., 2014; Mich et al., 2014), we found NSCs in quiescent and active states. Adding to this picture, our single-cell approach uncovered that during the progression from a quiescent to an active state, NSCs transit through multiple intermediate states that could not be captured with commonly used markers or population-based studies. Most interestingly, we find NSCs in a state of dormancy characterized by a particularly low protein synthesis rate. These cells progress to activation through an intermediate state of primed quiescence, in which cells begin to increase their protein synthesis machinery. These two states closely resemble the recently described transition from $G_0$ to an enhanced $G_{alert}$ quiescent state in muscle satellite cells (Rodgers et al., 2014). It will be interesting to study whether this constitutes a general mechanism of stem cell activation in homeostasis across many organs. Our data point toward a prominent role for translational control during several steps of NSC lineage progression. As more experimental tools become available to study protein synthesis in vivo (Buszczak et al., 2014), it will be important to study whether translational control exerts a regulatory effect on specific sets of transcripts to control NSC behavior.

One key step for future functional studies on NSCs is to find stage-specific markers to distinguish them from other niche cells. Our comparative analysis of the transcriptome of individual NSCs and other related cell types, such as neuroblasts, TAPs, oligodendrocytes, and parenchymal astrocytes will help in the identification of NSC- and subtype-specific markers. As an example, we have found that CD9 expression levels distinguishe parenchymal astrocytes from SVZ NSCs. Of note, high CD9 also marks Prom1$^{-}$ GLAST-expressing cells exclusively in the SVZ and may thus allow the identification of a broader pool of NSCs than the one marked by Prom1. In addition, stage-specific markers will allow distinction of the different subpopulations within NSCs, which will be critical to address the kinetics and the neurogenic contribution of these cells. Previous fate mapping of Glast-expressing cells demonstrates that Glast$^{high}$ cells, presumably including dormant NSCs, are indeed neurogenic during niche repopulation (Mich et al., 2014). In line with this finding, we show that after an ischemic insult NSCs transition from a dormant to a primed-quiescent state, shown to be neurogenic in transplantation experiments. However, definitive proof of their neurogenic ability awaits future confirmation by transplantation studies or exclusive lineage tracing of dormant NSCs.

Single-cell analysis also enabled us to study the coordinated expression of subtype-specific TFs in NSCs at higher throughput than conventional methods. Such analyses will provide a powerful means to identify TFs involved in the specification of particular neuronal subtypes that may be used for reprogramming approaches (Amamoto and Arlotta, 2014).

The high degree of NSC heterogeneity found in our study also poses some important questions. Do the different subgroups contribute equally to homeostatic neurogenesis? Do some cells serve as a reservoir for brain repair? Along this line, we have found here that dormant and quiescent NSCs are more responsive to an injured environment. This is consistent with the notion that qNSCs express a higher number of membrane receptors to sense and respond to changes in their microenvironment (Codega et al., 2014). However, within the injury niche, only a subset of NSCs responds to IFN-$\gamma$. The nature of this heterogeneous response remains unknown, and we can only speculate that it might be influenced by temporal dynamics, position within the niche, or paracrine inhibition from early responder cells (Shalek et al., 2014). Likewise, the presence of the injury-signature in some *Ifngr1*$^{-/-}$ NSCs might be explained by the concomitant activity of other cytokines. Interestingly, IFN-$\gamma$ promotes exit from dormancy into a primed quiescent state. A similar role of interferon signaling has been previously reported for hematopoietic stem cells where an acute treatment with interferon-$\alpha$ triggers proliferation of dormant HSCs (Essers et al., 2009).

In summary, our data show that multiple molecularly distinct groups of NSCs that respond differently to physiological stimuli coexist in the adult neurogenic niche. We anticipate that the understanding and implementation of this molecular diversity will be critical in harnessing the potential of NSCs for brain repair.

## EXPERIMENTAL PROCEDURES

### Animals

C57BL/6 male mice between 8 and 12 weeks of age were used for all experiments. Mice were purchased from Charles River or bred in house at the DKFZ Center for Preclinical Research facility. Animals were housed under standard conditions and fed ad libitum. Male and female *Ifngr1*$^{-/-}$ (*Ifnar1*$^{+/-}$ and *Ifnar1*$^{-/-}$) (Huang et al., 1993) mice were 3 months old and backcrossed to a C57BL/6 background. All procedures were in accordance with the DKFZ guidelines and approved by the Regierungpräsidium Karlsruhe.

### Global Forebrain Ischemia

Global ischemia was performed using the bilateral common carotid artery occlusion method essentially as previously described (Yoshioka et al., 2011). Briefly, mice were deeply anesthetized and a laser Doppler probe (Perimed) was fixed at the skull for continuous measurement of regional cerebral blood flow (rCBF). Next, both common carotid arteries were exposed and clamped with surgical microclamps (Stoelting) for a period of 22 min prior to reperfusion. Only animals with an ischemic rCBF smaller than 13% of the pre-ischemic value were used.

### Stereotactic Injections

For transplantation experiments, two injections containing approximately 2,000–4,000 cells purified by flow cytometry from pan-YFP or pan-Tomato (Rainbow 2) mice were performed at the following coordinates zeroed at the bregma: 0.0/1.4/−2.1 and 0.5/1.2/−2.2 mm. Recipient mice were age and gender matched. Mice were sacrificed at the indicated time point. To measure protein synthesis in the SVZ, 1.5 μl of 20mM OP-Puro (dissolved in 5% DMSO in PBS) was injected into the lateral ventricle at the following coordinates: −0.5/1.2/−2 mm. Mice were sacrificed 30 min later and the contralateral hemisphere was analyzed.

### Flow Cytometry and Cell Picking

Animals were sacrificed by cervical dislocation and their brains were immediately placed on ice. The lateral SVZ was microdissected as a whole mount as previously described (Mirzadeh et al., 2010). Pooled SVZ tissue (one to five mice) was digested with trypsin and DNase according to the Neural Tissue Dissociation kit in a Gentle MACS Dissociator (Miltenyi). To isolate cortical and striatal astrocytes, we microdissected tissue pieces similarly sized to the SVZ wholemount from the cortex immediately above the corpus callosum and from the striatum after having dissected out the ventricle wall, and we processed them in parallel. After being sorted, cells were manually picked with a micropipette under continuous visual inspection in a Zeiss microscope. Picked cells were frozen at −80°C until further processing. For details regarding cell staining, sorting, and picking, see Supplemental Experimental Procedures.

### Library Preparation and Sequencing

RNA-seq libraries were prepared using Smart-seq2 technology as previously described (Picelli et al., 2014). Briefly, lysed cells were thawed and subjected to reverse transcription using an oligo(dT) primer and a locked nucleic acid (LNA)-containing template-switching oligonucleotide (Exiqon). In a randomly selected set of samples, we included ERCC Spike-Ins (Ambion) at a 1:1,000,000 dilution. Full-length cDNAs were amplified by 20 cycles of PCR using KAPA HiFi DNA polymerase (KAPA biosystems). Amplified cDNAs for every single cell were quantified and run on a High Sensitivity Bioanalyzer chip (Agilent). Samples containing no material or in which the cDNA appeared degraded were excluded (approximately 10%) from further processing. cDNAs were then converted into libraries for Illumina sequencing according to the Nextera XT Sample Preparation (Illumina) protocol with minor modifications. For population studies, approximately 1 ng of total RNA from approximately 1,000 cells was subjected to the SMARTer Ultra Low RNA kit (Clontech) according to the manufacturer's instructions. Samples were sequenced in an Illumina HiSeq 2000. For details, see Supplemental Experimental Procedures.

### Immunohistochemistry, Immunocytochemistry, and Cell Culture

Mice were transcardially perfused first with HBSS and then with 4% paraformaldehyde (PFA). SVZ wholemounts were processed for staining as previously described (Mirzadeh et al., 2010). For staining of cells directly after sorting, cells were plated in PDL and laminin-coated Lab-Tek chambers (Thermo) until they adhered or until the duration of the respective treatment. Afterward, cells were fixed for 20 min with 2% PFA in the medium and stained following standard procedures. The alkyne group in OP-Puro was detected using an Alexa-488- or Alexa-647-coupled azide according to the Click-iT Cell Reaction Buffer Kit (Life Technologies). FAC-sorted cells were incubated in the absence of growth factors in Neurobasal medium supplemented with B27, glutamine, and Pen/Strep. Cells were treated with 10 μM DAPT (Sigma) or DMSO (Sigma) for the indicated time. OP-Puro (Jena Biosciences) was added to the cells at 50 μM for 1 hr prior to fixation. Antibodies are listed in the Supplemental Experimental Procedures.

### Imaging

Confocal images were acquired in a Leica TCS SP5 confocal microscope. Image J was used for processing and images were only adjusted for brightness and contrast. For OP-Puro quantifications, all images within compared groups were acquired with identical settings. Analysis was performed in ImageJ software using a custom-written macro for unbiased segmentation and quantification of pixel intensity and cell size. OP-Puro quantifications are depicted as the integrated pixel intensity within a cell and normalized to a reference group within the same experiment. In graphs, mean and SD are shown. Illustrator CS5 (Adobe) was used for figure assembly.

### Processing and Analysis of Single-Cell RNA-Seq Data

Data analysis was performed as described in detail in the Supplemental Experimental Procedures. Briefly, the quality of raw reads was checked by FASTQC and reads were trimmed using Btrim64 (Kong, 2011). Trimmed reads were mapped to mouse genome (ENSEMBL Release 78) using STAR_2.4.0g (Dobin et al., 2013). RNA-seq data quality metrics for each cell, including total reads, transcriptome mapped reads, and transcriptome mapped rate was calculated by picard-tools-1.123 (https://broadinstitute.github.io/picard/) as shown in Table S1. Next, gene expression matrices were generated as previously described (Shalek et al., 2013, 2014). To compare expression levels of different genes across samples, we performed an additional TMM (trimmed mean of M values) normalization on FPKM (fragments per kilobase of transcript per million mapped reads) using Trinity (Haas et al., 2013) (http://trinityrnaseq.github.io/) based on edgeR (Robinson and Oshlack, 2010). PCA on gene expression matrices was done using custom R scripts based on FactoMineR library (http://factominer.free.fr/). Hierarchical clustering was performed on cells and on the genes identified by PCA using Euclidean distance or correlation metric. We also assessed the stability of the number of clusters identified by hierarchical clustering by a previously described resampling method (Wu et al., 2014). To reconstruct the pseudotemporal order of NSCs, we applied Monocle (Trapnell et al., 2014) to the expression pattern of genes with the highest coordinates ($p < 10e-5$) with the first three dimensions of PCA. To determine qNSC1-enriched genes, a likelihood-ratio test (McDavid et al., 2013) was used to test genes that were differentially expressed between qNSC1 and astrocytes ($p < 0.01$, Benjamini and Hochberg corrected).

## ACCESSION NUMBERS

Data have been deposited in GEO under accession number GEO: GSE67833.

## SUPPLEMENTAL INFORMATION

Supplemental Information for this article includes five figures, four tables, and Supplemental Experimental Procedures and can be found with this article online at http://dx.doi.org/10.1016/j.stem.2015.07.002.

## AUTHOR CONTRIBUTIONS

E.L.-B. designed the study, performed experiments, analyzed and interpreted data, and prepared the manuscript. S.Z. participated in experimental design, performed the computational analyses, interpreted data, and prepared the manuscript. A.B. performed experiments and analyzed data. G.S.-C. and K.Z. performed experiments. A.M.-V. aided in conceptual design and study coordination, interpreted data, and prepared the manuscript.

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